

The Multi-Dimensional Quantitative Risk Framework (MQRF)

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Traditional risk frameworks rely on qualitative matrices such as the 3×3 risk matrix.

Problems:

- **Range Compression:** Different risks collapse into the same category.
- **Linear Fallacy:** Small losses and catastrophic events may receive the same score.
- **Black Swan Blindness:** Low probability but catastrophic events are underestimated.

MQRF introduces a **non-linear quantitative risk framework**.

Master Risk Determination Formula

The framework computes the **Residual Risk Score (RS)**.

$$RS = \left[\frac{I^{\alpha} \times L \times V^{\beta}}{D} \right] \times (1 - CE)$$

Where:

- I = Impact
- L = Likelihood
- V = Velocity
- D = Detectability
- CE = Control Effectiveness
- α = Impact sensitivity exponent
- β = Velocity sensitivity exponent

Impact Modeling

Impact is normalized within the range:

$$0 \leq I \leq 10$$

Impact amplification is controlled by the exponent α .

$$I^\alpha$$

Recommended value:

$$\alpha = 1.2$$

Example:

$$10^{1.2} = 15.84$$

This ensures catastrophic risks receive higher weight.

Impact Pillar Normalization

Impact is computed as a weighted sum:

$$I = 0.4R + 0.3S + 0.2F + 0.1O$$

Where:

- R = Regulatory Impact
- S = Strategic Impact
- F = Financial Impact
- O = Operational Impact

$$\sum w_i = 1$$

Weights can be adjusted depending on organizational priorities.

Impact Calibration Guidelines (IIIT Hyderabad)

Impact score I is computed using institutional pillars:

$$I = 0.4R + 0.3S + 0.2F + 0.1O$$

Score	Financial (F)	Operational (O)	Regulatory (R)	Strategic (S)
0–2	Negligible (< \$1k)	< 1 hr delay	No legal issue	Localized gossip
3–5	Moderate (< \$50k)	1–8 hr outage	Minor fine / penalty	Regional news
6–8	Severe (< \$500k)	1–3 day outage	Lawsuit / inquiry	National PR crisis
9–10	Catastrophic (> \$1M)	Total shutdown	License revocation	Irreparable brand damage

Likelihood Modeling: Stochastic Events

For natural or random events, likelihood is computed using a Poisson-style model:

$$L = 1 - e^{-\lambda F_h}$$

Where:

- F_h = historical event frequency (events per year)
- $\lambda = \frac{\ln(2)}{F_{target}}$
- F_{target} = frequency considered “Likely” ($L = 0.5$)

Example Calibration

Assume the institution defines

$$F_{target} = 5 \text{ events/year}$$

Then

$$\lambda = \frac{\ln(2)}{5} = 0.1386$$

Example Evaluations

Using

$$L = 1 - e^{-0.1386F_h}$$

- If $F_h = 1$ event/year:

$$L = 1 - e^{-0.1386} = 0.13$$

- If $F_h = 5$ events/year:

$$L = 1 - e^{-0.693} = 0.50$$

- If $F_h = 10$ events/year:

$$L = 1 - e^{-1.386} = 0.75$$

Interpretation

- Rare events produce small likelihood values
- Frequent events rapidly approach $L \rightarrow 1$

Likelihood Modeling (Adversarial)

For human adversaries:

$$L = \max(0.05, M' \times E')$$

Where:

- M' = attacker motivation
- E' = exposure / attack surface

Interpretation:

If either M' or E' approaches zero, overall likelihood approaches zero.

Velocity Modeling

Velocity measures the **time-to-impact (TTI)**.

$$1.0 \leq V \leq 2.0$$

Where:

- $V = 1.0$: Slow risks (years)
- $V = 1.5$: Medium risks (days/weeks)
- $V = 2.0$: Immediate risks (seconds)

Velocity influence is controlled by exponent β :

$$V^\beta$$

Recommended:

$$1.0 \leq \beta \leq 1.5$$

Detectability

$$1 \leq D \leq 5$$

Higher detectability reduces risk.

- 1 = Blind
- 3 = Logged monitoring
- 5 = Predictive AI detection

Control Effectiveness

$$0 \leq CE < 1$$

Residual risk:

$$RS = R_{raw}(1 - CE)$$

Velocity and Detectability Calibration

Velocity measures **Time To Impact (TTI)**.

Detectability measures the **visibility of monitoring systems**.

Score	Velocity (V) – TTI	Detectability (D) – Visibility
1.0	Years (Policy / Brand shift)	Blind: No monitoring or logs exist
1.5	Weeks / Days (Process failure)	Manual: Found during periodic audits
2.0	Seconds (Cyber / Terror attack)	Predictive: AI anomaly detection

Control Effectiveness (CE) Maturity Model

Control effectiveness represents the percentage of risk mitigated by safeguards.

$$0 \leq CE < 1$$

Value	Maturity Level	Description
0.0 – 0.1	Ad-Hoc	No formal process. Safety depends on individual heroics or luck.
0.2 – 0.4	Initial	Processes documented but inconsistent. No regular testing.
0.5 – 0.7	Defined	Standardized processes across the institution. Occasional audits.
0.8 – 0.9	Managed	Controls automated and monitored. KPIs exist.
0.95+	Optimized	Continuous automated testing. Near-perfect reliability (e.g., Tier-4 Data Centers).

Case Study: Library ID Abuse

$$I = 1.5$$

$$I^{\alpha} = 1.5^{1.2} = 1.63$$

$$L = 0.24$$

$$V^{\beta} = 1.1^{1.2} = 1.12$$

$$RS = \left[\frac{1.63 \times 0.24 \times 1.12}{2} \right] \times 0.7 = 0.15$$

Negligible Risk

Case Study: Earthquake at IIT Roorkee

$$I = 8.5$$

$$I^\alpha = 8.5^{1.2} = 13$$

$$L = 0.5 \text{ (0.29 for IIIT Hyderabad)}$$

$$V^\beta = 2^{1.2} = 2.30$$

$$RS = \left[\frac{13 \times 0.5 \times 2.30}{1} \right] \times 0.8 = 11.96$$

Severe Risk

Case Study: Ransomware Attack on Government

$$I = 10$$

$$I^{\alpha} = 10^{1.2} = 15.84$$

$$L = 0.72$$

$$V^{\beta} = 2^{1.2} = 2.30$$

$$RS = \left[\frac{15.84 \times 0.72 \times 2.30}{1} \right] \times 0.85 = 22.28$$

Critical Risk

Risk Interpretation

0 – 1	Acceptable
1 – 5	Moderate
5 – 15	Severe
> 15	Critical

Conclusion

MQRF provides:

- Nonlinear risk amplification
- Explicit modeling of catastrophic impact
- Quantitative decision support